Jackson Liao-Cheng
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EDUCATION

Cornell University (Duffield College of Engineering) **Expected Graduation: May 2028**
- B.S., Computer Science & Operations Research **Cumulative GPA: 3.96**

EXPERIENCE

Google Ithaca, NY
Student Software Developer July 2026 - Present
- Trained and deployed a PyTorch R3D-18 video classification model on ~150,000 clips, using Kubernetes to generate GPU-intensive distributed ML workloads for realistic failure simulation
- Reduced ML failure diagnosis time by 50% via a Go service correlating GPU, CPU, and database telemetry
- Improved root-cause identification accuracy 30% through semantic retrieval of past incidents via AlloyDB
- Unified 15+ observability metrics linking ML performance to infra/database health with Grafana dashboards

Splunk (A Cisco Company) Boulder, CO
Software Engineer Intern May 2026 - Aug 2026
- Automated new feature cloud spend monitoring to eliminate multi-day manual review, saving 100 hours monthly and scaling new feature coverage by 2x via LangGraph workflows and MCP servers
- Mitigated model hallucinations via RAG chains and HITL, blocking 100% of invalid outputs, and prevented automation error with comprehensive unit, end-to-end, and user acceptance testing
- Developed and evaluated an ML model to group EC2 spend by risk, with 80% prediction accuracy
- Applied risk groups to size savings plans, saving ~$33,000 monthly at 93% commitment utilization
- Built an AWS discount calculator to view per-stack costs under several plans to guide decisions
- Worked in an Agile environment, participating in iterative development, testing, and code reviews

Cornell University Ithaca, NY
Undergraduate Researcher Jan 2026 - June 2026
- Conducted research on queuing network optimization through deep reinforcement learning

American Red Cross Buffalo, NY
Software Engineer Intern June 2024 - Aug 2024
- Developed utility script with a REST API to archive 25,000 Jira issues, saving 200+ dev hours
- Engineered and documented a Flask-based web application to trigger and monitor archival jobs
- Collaborated with engineers on internal tooling supporting migration to Kubernetes on AWS

PROJECTS

Resume Tailorer | *Python, TypeScript, FastAPI, React, PostgreSQL* Nov 2025 - Feb 2026
- Designed and implemented end-to-end resume parsing and job-based tailoring application
- Engineered LLM pipeline with JSON-schema prompts to convert model text into structured data
- Automated resume-to-job matching workflows, generating role-specific, AI-driven suggestions
- Developed interactive frontend to find jobs and review AI-generated resume recommendations

Critterworld | *Java, Spark, Gradle, JUnit, Git* Sept 2025 - Dec 2025
- With a team, architected and implemented a parser and AST for a custom critter behavior language
- Applied the visitor design pattern to support AST mutations for evolutionary simulation features
- Engineered the core simulation model, including critter logic, rule evaluation, and world updates
- Built a GUI that renders and simulates worlds, using multithreading to maintain responsiveness

TECHNICAL SKILLS

Languages: Python, Java, Go, JavaScript/TypeScript, OCaml, SQL, R, HTML/CSS
Backend & Databases: REST, Flask, FastAPI, PostgreSQL, AlloyDB
Cloud & Infrastructure: AWS, GCP, Kubernetes, Docker, Linux, Git
AI/ML: PyTorch, RAG, LangChain/LangGraph, MCP, Vector Databases, Model Evaluation